Data Manipulation Language (DML) & Transactions

Course: DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
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1. CRUD Statements on Core & Junction Relations

```
-- INSERT Case, Person, and Bridge Table Association
INSERT INTO cases (case_id, title, description, case_type, status, priority, lead_officer_id)
VALUES ('CASE-2026-901', 'Operation Cobalt Wire', 'Offshore satellite telemetry burst', 'Electronic Surveillance', 'Open', 'Critical

INSERT INTO people (person_id, full_name, category, contact_phone, address)
VALUES ('PER-901', 'Farhan Qureshi', 'Subject', '+91-98450-99881', 'Mangaluru Port Dock 4B');

INSERT INTO case_people (case_id, person_id, role_in_case)
VALUES ('CASE-2026-901', 'PER-901', 'Chief Systems Operator');

-- UPDATE Status and Evidence Vault Location
UPDATE cases SET status = 'Under Investigation', priority = 'Critical' WHERE case_id = 'CASE-2026-901';
UPDATE evidence SET storage_reference = 'Cyber Forensics Station 2' WHERE evidence_id = 'EVD-901';

-- DELETE Association (Decouples suspect while preserving master person entity)
DELETE FROM case_people WHERE case_id = 'CASE-2026-901' AND person_id = 'PER-901';
```

2. ACID Transaction with SAVEPOINT & Recovery Demonstration

```
BEGIN;
-- Step 1: Register exhibit
INSERT INTO evidence (evidence_id, case_id, title, evidence_type, storage_reference)
VALUES ('EVD-902', 'CASE-2026-901', 'Encrypted MicroSD Card', 'Digital Storage', 'Vault Tray 9');

-- Step 2: Establish savepoint
SAVEPOINT after_evidence_created;

-- Step 3: Attempt assignment to non-existent officer (Validation error)
-- Trigger rollback to savepoint:
ROLLBACK TO SAVEPOINT after_evidence_created;

-- Step 4: Correct assignment with verified officer
INSERT INTO evidence_custody (evidence_id, officer_id, event_type, notes)
VALUES ('EVD-902', 'OFF-001', 'Initial Vault Deposition', 'Logged after savepoint recovery.');

COMMIT;
-- VERIFICATION QUERY:
-- SELECT evidence_id, title, storage_reference FROM evidence WHERE evidence_id = 'EVD-902';
-- OUTPUT: EVD-902 | Encrypted MicroSD Card | Vault Tray 9 (Transaction committed cleanly)
```